JQuery_Day11_Hands On-Chintha Ramkrishna

Problem 1 :-

How does this Student Feedback Form validate user input using jQuery and display a success message after submission?

Code :-

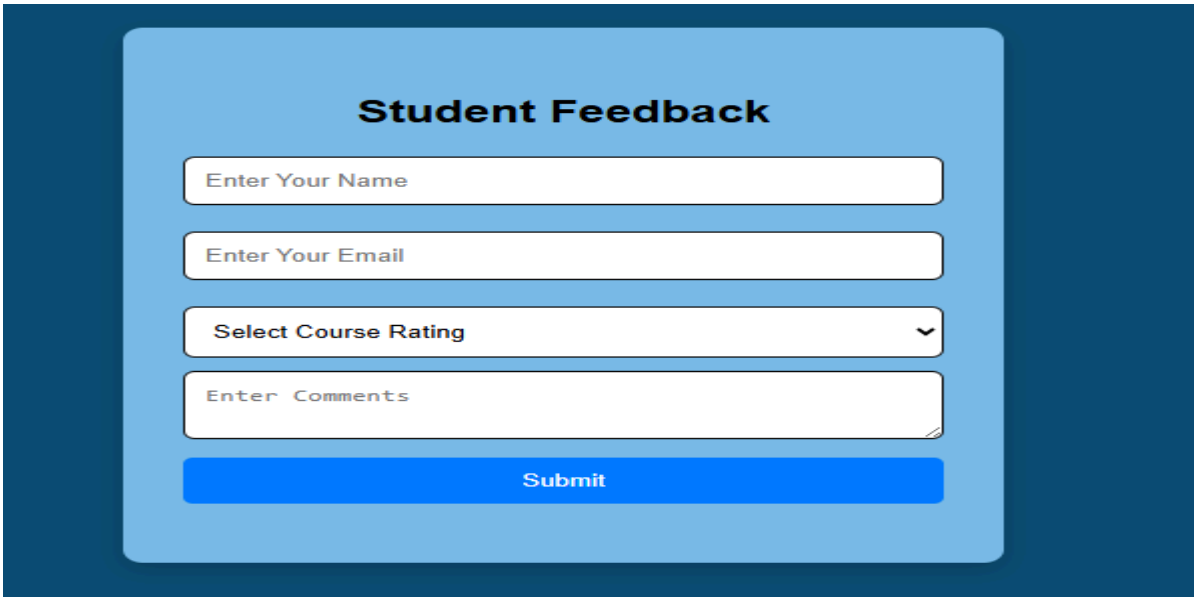
```html
<!DOCTYPE html>
<html>
<head>
    <title>Student Feedback Form</title>

    <!-- Technical Constraint: jQuery CDN -->
    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>

    <style>
        body{
            font-family: Arial, sans-serif;
            background:#0e4e74;
            display:flex;
            justify-content:center;
            align-items:center;
            height:100vh;
            margin:0;
        }

        .form-box{
            background:rgb(122, 188, 231);
            padding:30px;
            width:380px;
            border-radius:10px;
            box-shadow:0px 5px 15px rgba(0,0,0,0.2);
        }

        h2{
            text-align:center;
            margin-bottom:20px;
        }

        input, select, textarea{
            width:100%;
            padding:10px;
```

```
            margin-bottom:10px;
            border-radius:6px;
            border:1px solid #0c0c0c;
            box-sizing:border-box;
        }

        .error{
            color:red;
            font-size:13px;
            margin-bottom:10px;
        }

        .success{
            color:green;
            margin-top:15px;
            text-align:center;
            font-weight:bold;
        }

        button{
            width:100%;
            padding:10px;
            background:#007bff;
            color:white;
            border:none;
            border-radius:5px;
            cursor:pointer;
        }

        button:hover{
            background:#0056b3;
        }
    </style>
</head>

<body>

<div class="form-box">

    <h2>Student Feedback</h2>

    <!-- Requirement: Feedback Form Fields -->
    <form id="feedbackForm">
```

```
        <input type="text" id="name" placeholder="Enter Your Name">
        <div id="nameError" class="error"></div>

        <input type="email" id="email" placeholder="Enter Your Email">
        <div id="emailError" class="error"></div>

        <select id="rating">
            <option value="">Select Course Rating</option>
            <option>Excellent</option>
            <option>Good</option>
            <option>Average</option>
            <option>Poor</option>
        </select>

        <textarea id="comments" placeholder="Enter
Comments"></textarea>

        <button type="button" id="submitBtn">Submit</button>

    </form>

    <!-- Requirement: Success message below form -->
    <div id="successMessage" class="success"></div>

</div>

<script>

$(document).ready(function(){

    // Event Handling using jQuery
    $("#submitBtn").click(function(){

        // Capture inputs using jQuery selectors
        let name = $("#name").val().trim();
        let email = $("#email").val().trim();

        // Clear old messages
        $(".error").text("");
        $("#successMessage").text("");

        // Validation
```

```
        if(name === ""){
            $("#nameError").text("Please enter your name.");
        }
        else if(email === ""){
            $("#emailError").text("Please enter your email.");

        }
        else{

            // Display success message dynamically
            $("#successMessage")
                .text("Thank you " + name + "! Your feedback has been
submitted successfully.");

            // Clear form fields
            $("#feedbackForm")[0].reset();
        }

    });

});

</script>

</body>
</html>
```

Output :-



Explanation :-

1. The form collects user input such as name, email, rating, and comments using HTML fields.

2. jQuery is used to handle the button click event with the `click()` function.

3. The program checks if the name and email fields are empty using simple `if` conditions.

4. If any required field is empty, an error message is displayed below that input field.

5. If all validations pass, a success message is shown dynamically and the form is reset.

Problem 2 :-

How does the Dynamic FAQ Toggle Section use jQuery to show and hide answers with animation?

Code:-

```
<!--Problem 2 (Level-1): Dynamic FAQ Toggle Section
Scenario:
A website contains a Frequently Asked Questions (FAQ) section. The
company wants answers to appear and disappear when users click on the
questions.
Requirements
• Create at least 5 FAQ questions with hidden answers.
• Use jQuery click() event to toggle answers.
• Apply slideToggle() or show()/hide() methods.
• Highlight the selected question using CSS class manipulation.

 Technical Constraints
• Use only jQuery methods for DOM manipulation.
• Use class selectors and ID selectors appropriately.
• Avoid inline JavaScript.

Expectations:
• Smooth UI interaction.
• Proper use of jQuery effects.
```

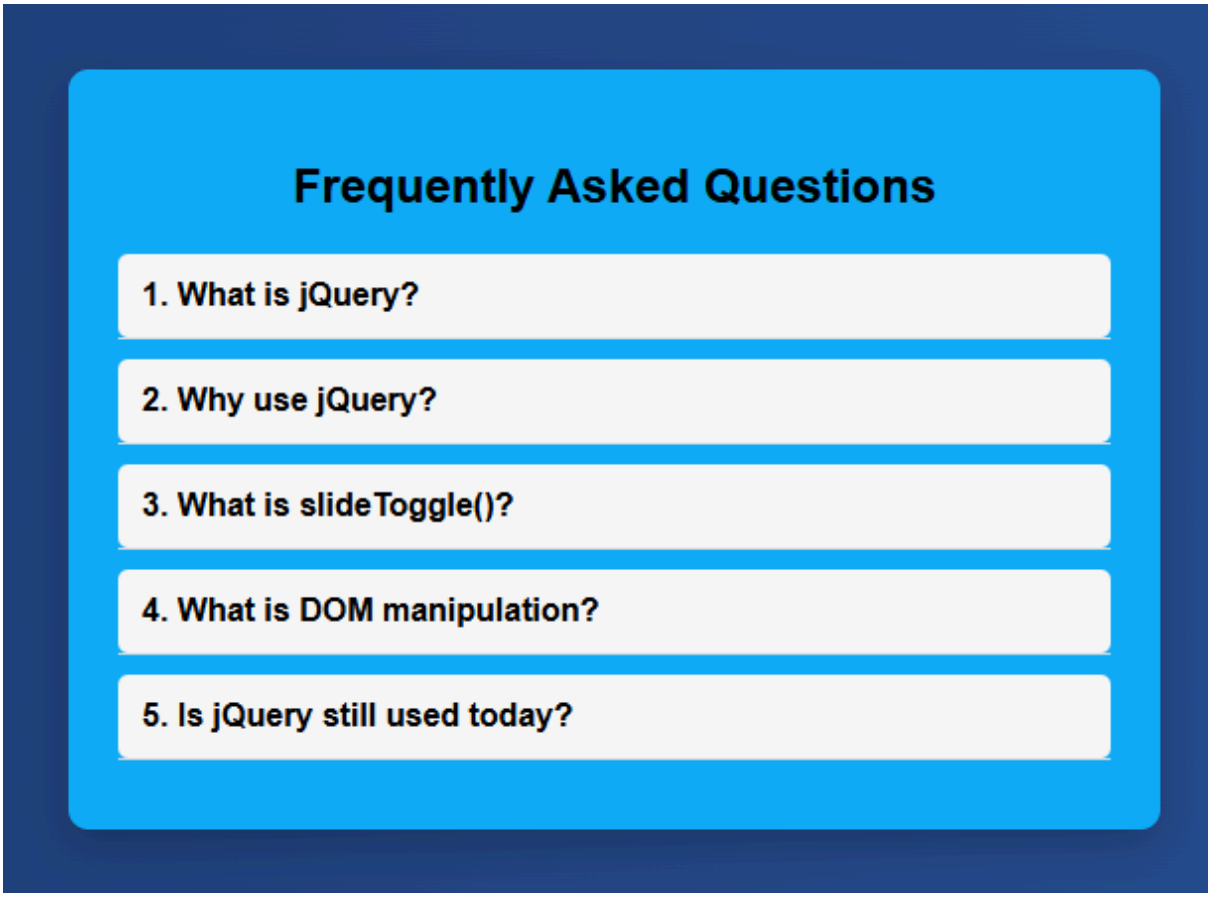
```
• Organized HTML structure.

Learning Outcome
• Implement event handling using jQuery.
• Use basic jQuery effects.
• Practice DOM traversal and manipulation.-->
<!DOCTYPE html>
<html>
<head>
    <title>FAQ Toggle Section</title>

    <!-- jQuery CDN -->
    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>

    <style>
        body{
            font-family: Arial, sans-serif;
            background: linear-gradient(to right, #1e3c72, #2a5298);
            display:flex;
            justify-content:center;
            align-items:center;
            height:100vh;
            margin:0;
        }

        .faq-container{
            width:500px;
            background:rgb(17, 173, 245);
            padding:25px;
            border-radius:10px;
            box-shadow:0px 8px 20px rgba(0,0,0,0.2);
        }

        h2{
            text-align:center;
            margin-bottom:20px;
        }

        .faq-item{
            margin-bottom:10px;
            border-bottom:1px solid #ddd;
        }
```

```
        .question{
            padding:12px;
            cursor:pointer;
            font-weight:bold;
            background:#f5f5f5;
            border-radius:5px;
        }

        .question:hover{
            background:#e0e0e0;
        }

        .answer{
            display:none;   /* Hidden initially */
            padding:10px;
            font-size:14px;
        }

        .active{
            background:#d909f5;
            color:white;
        }

    </style>
</head>

<body>

<div class="faq-container">
    <h2>Frequently Asked Questions</h2>

    <!-- FAQ 1 -->
    <div class="faq-item">
        <div class="question">1. What is jQuery?</div>
        <div class="answer">jQuery is a JavaScript library used to
simplify DOM manipulation and event handling.</div>
    </div>

    <!-- FAQ 2 -->
    <div class="faq-item">
        <div class="question">2. Why use jQuery?</div>
        <div class="answer">It reduces code and provides easy-to-use
methods for animations and AJAX.</div>
```

```
    </div>

    <!-- FAQ 3 -->
    <div class="faq-item">
        <div class="question">3. What is slideToggle()?</div>
        <div class="answer">slideToggle() shows or hides content with a
sliding animation effect.</div>
    </div>

    <!-- FAQ 4 -->
    <div class="faq-item">
        <div class="question">4. What is DOM manipulation?</div>
        <div class="answer">DOM manipulation means changing HTML
content dynamically using JavaScript or jQuery.</div>
    </div>

    <!-- FAQ 5 -->
    <div class="faq-item">
        <div class="question">5. Is jQuery still used today?</div>
        <div class="answer">Yes, it is still used in legacy systems and
small projects.</div>
    </div>

</div>

<script>

$(document).ready(function(){

    $(".question").click(function(){

        // Remove active class from other questions
        $(".question").not(this).removeClass("active");

        // Toggle active class for clicked question
        $(this).toggleClass("active");

        // Hide other answers
        $(".answer").not($(this).next()).slideUp();

        // Toggle clicked answer
        $(this).next(".answer").slideToggle();
```

```
    });

});

</script>

</body>
</html>
```

Output :-



Explanation:-

1. The FAQ section contains five questions, and each answer is hidden initially using `display: none` in CSS.

2. jQuery `click()` event is used to detect when a user clicks on a question.

3. The `slideToggle()` method smoothly shows or hides the selected answer with an animation effect.

4. The `removeClass()` and `toggleClass()` methods are used to highlight the currently selected question.

5. The `slideUp()` method hides other answers, ensuring only one answer is visible at a time for better UI interaction.

Problem 3 :-
How does the Product Cart Counter update the cart count dynamically using jQuery event delegation?

Code:-

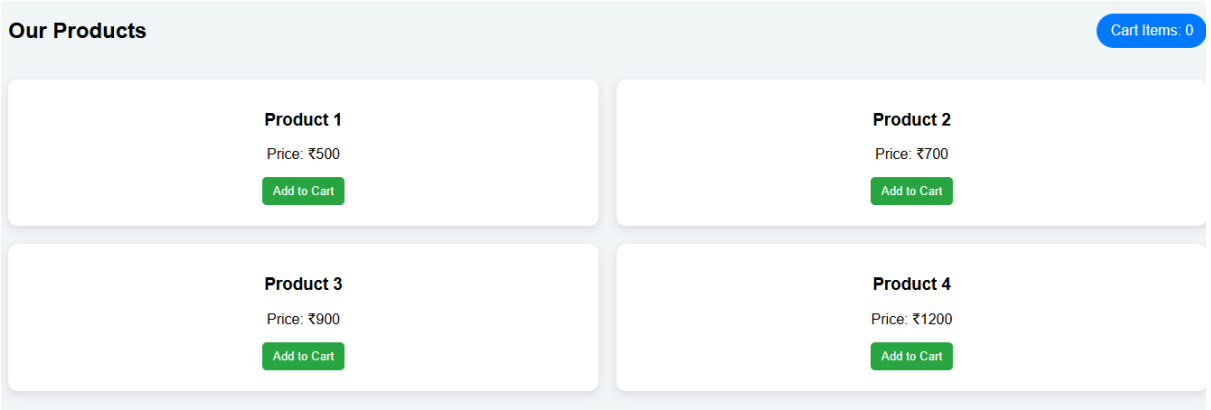
```
<!DOCTYPE html>
<html>
<head>
    <title>Product Cart Counter</title>

    <!-- jQuery CDN -->
    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>

    <style>
        body{
            font-family: Arial, sans-serif;
            background:#f4f6f9;
            margin:0;
            padding:20px;
        }

        .header{
            display:flex;
            justify-content:space-between;
            align-items:center;
            margin-bottom:20px;
        }

        .cart{
            background:#007bff;
            color:white;
            padding:10px 15px;
            border-radius:20px;
        }

        .products{
```

```
            display:grid;
            grid-template-columns:repeat(2, 1fr);
            gap:20px;
        }

        .product{
            background:white;
            padding:15px;
            border-radius:10px;
            box-shadow:0px 4px 10px rgba(0,0,0,0.1);
            text-align:center;
        }

        button{
            padding:8px 12px;
            background:#28a745;
            color:white;
            border:none;
            border-radius:5px;
            cursor:pointer;
        }

        button:disabled{
            background:gray;
            cursor:not-allowed;
        }

        .added-msg{
            color:green;
            font-size:13px;
            margin-top:8px;
        }

    </style>
</head>

<body>

<div class="header">
    <h2>Our Products</h2>
    <div class="cart">
        Cart Items: <span id="cartCount">0</span>
    </div>
```

```
</div>

<div class="products">

    <div class="product">
        <h3>Product 1</h3>
        <p>Price: ₹500</p>
        <button class="addBtn" data-name="Product 1">Add to
Cart</button>
        <div class="added-msg"></div>
    </div>

    <div class="product">
        <h3>Product 2</h3>
        <p>Price: ₹700</p>
        <button class="addBtn" data-name="Product 2">Add to
Cart</button>
        <div class="added-msg"></div>
    </div>

    <div class="product">
        <h3>Product 3</h3>
        <p>Price: ₹900</p>
        <button class="addBtn" data-name="Product 3">Add to
Cart</button>
        <div class="added-msg"></div>
    </div>

    <div class="product">
        <h3>Product 4</h3>
        <p>Price: ₹1200</p>
        <button class="addBtn" data-name="Product 4">Add to
Cart</button>
        <div class="added-msg"></div>
    </div>

</div>

<script>

$(document).ready(function(){

    let count = 0;
```

```
    // Event Delegation
    $(".products").on("click", ".addBtn", function(){

        count++;

        // Update cart counter
        $("#cartCount").text(count);

        // Disable button
        $(this).prop("disabled", true);

        // Change button text
        $(this).text("Added");

        // Add custom attribute
        $(this).attr("data-added", "true");

        // Show confirmation message
        $(this).siblings(".added-msg")
               .text("Product added successfully!");

    });

});

</script>

</body>
</html>
```

Output :-



Explanation :-

1. The program displays multiple products, each with an "Add to Cart" button and a cart counter at the top.

2. jQuery event delegation is implemented using `.on("click", ".addBtn", function())` on the parent `.products` container.

3. When a button is clicked, the cart counter increases and updates dynamically using `text()`.

4. The clicked button is disabled using `prop("disabled", true)` and its text is changed to "Added".

5. A confirmation message is displayed below the product using `siblings()` to improve user interaction.

Problem 4 :-
How does the Interactive To-Do List manage adding, completing, and deleting tasks using jQuery?

Code :-

```
<!DOCTYPE html>

<html>

<head>

    <title>Interactive To-Do List</title>



    <!-- jQuery CDN -->

    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>



    <style>

        body{

            font-family: Arial, sans-serif;
```

```
        background:#1d2163;
        display:flex;
        justify-content:center;
        align-items:center;
        height:100vh;
        margin:0;
    }

    .todo-box{
        background:rgb(55, 88, 104);
        padding:30px;
        width:400px;
        border-radius:10px;
        box-shadow:0px 5px 15px rgba(0,0,0,0.2);
    }

    h2{
        text-align:center;
        margin-bottom:20px;
    }

    .input-group{
        display:flex;
        gap:10px;
    }
```

```
input{
    flex:1;
    padding:10px;
    border-radius:6px;
    border:1px solid #ccc;
}

button{
    padding:10px 15px;
    background:#007bff;
    color:white;
    border:none;
    border-radius:6px;
    cursor:pointer;
}

button:hover{
    background:#0056b3;
}

ul{
    list-style:none;
    padding:0;
    margin-top:20px;
```

```
        }


        li{
            display:flex;
            justify-content:space-between;
            align-items:center;
            padding:8px;
            background:#f1f1f1;
            margin-bottom:8px;
            border-radius:5px;
        }


        .completed{
            text-decoration: line-through;
            color:gray;
        }


        .delete-btn{
            background:red;
            padding:5px 8px;
            font-size:12px;
        }


        .stats{
            margin-top:15px;
```

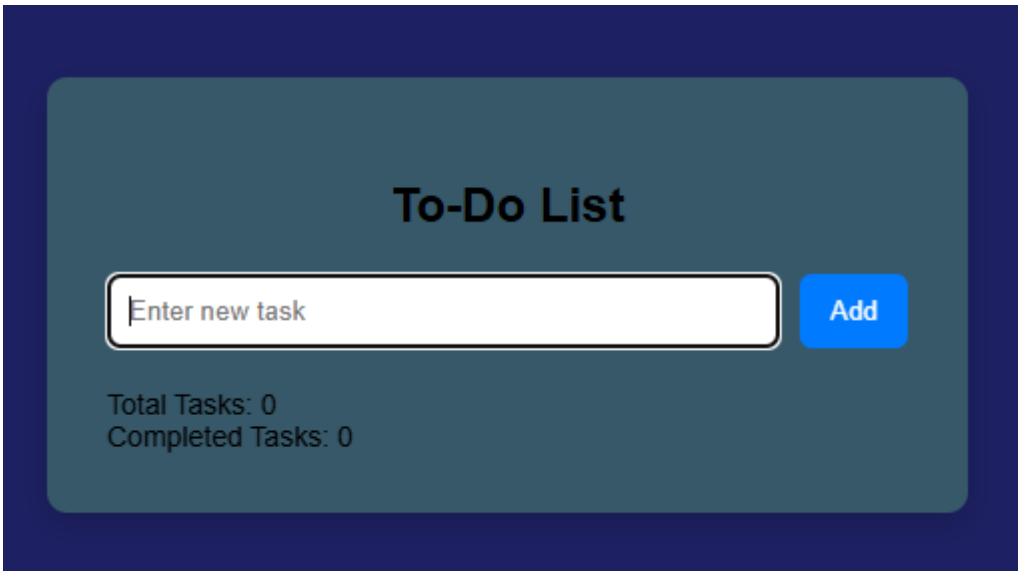
```
            font-size:14px;

        }

    </style>

</head>


<body>


<div class="todo-box">


    <h2>To-Do List</h2>


    <!-- Input + Add Button -->

    <div class="input-group">

        <input type="text" id="taskInput" placeholder="Enter new task">

        <button id="addTask">Add</button>

    </div>


    <!-- Task List -->

    <ul id="taskList"></ul>


    <!-- Task Count -->

    <div class="stats">

        Total Tasks: <span id="totalCount">0</span><br>

        Completed Tasks: <span id="completedCount">0</span>

    </div>
```

```
</div>

<script>

$(document).ready(function(){

    function updateCounts(){
        let total = $("#taskList li").length;
        let completed = $("#taskList li.completed").length;

        $("#totalCount").text(total);
        $("#completedCount").text(completed);
    }

    // Add Task
    $("#addTask").click(function(){

        let taskText = $("#taskInput").val().trim();

        if(taskText !== ""){

            $("#taskList").append(
                "<li>" +
                "<span class='task-text'>" + taskText + "</span>" +
```

```
                "<button class='delete-btn'>Delete</button>" +
                "</li>"
            );

            $("#taskInput").val("");
            updateCounts();
        }
    });

    // Mark as Completed (Dynamic Event Handling)
    $("#taskList").on("click", ".task-text", function(){
        $(this).parent().toggleClass("completed");
        updateCounts();
    });

    // Delete Task
    $("#taskList").on("click", ".delete-btn", function(){
        $(this).parent().remove();
        updateCounts();
    });

});

</script>
```

```
</body>

</html>
```

Output :-



Explanation :-

1. When the user enters a task and clicks the "Add" button, jQuery appends a new `<li>` element dynamically to the task list.

2. The `updateCounts()` function calculates total and completed tasks using selectors like `#taskList li` and updates the counters.

3. Event delegation is used with `.on()` to handle clicks on dynamically created task elements.

4. Clicking on the task text toggles the `completed` class, which applies a line-through style using CSS.

5. Clicking the "Delete" button removes the task from the list and updates the task statistics automatically.